App Development Project Report
Trekking Management Application

1. Student Details

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I am pursuing the IIT Madras BS Degree in Data Science and Applications along with a B.Tech in Information Technology. This project focuses on managing trekking operations, bookings, trek staff activities, notifications, reports, and user interactions through a role-based web application.

2. Project Details

Project Title: Trekking Management Application

Problem Statement:
Design and develop a trekking management platform that allows administrators, trek staff, and trekkers to efficiently manage trekking events, registrations, bookings, schedules, notifications, and reports.

Approach:
• Flask REST backend with role-based authentication
• SQLite database with SQLAlchemy ORM
• Vue.js frontend for interactive dashboards
• JWT-based authentication and authorization
• Trek management, booking management, and staff operations
• Export reports, reminders, notifications, and scheduled background jobs

3. AI/LLM Declaration

ChatGPT (GPT-5) was used for selected code suggestions, debugging support, API design discussions, documentation generation, and frontend planning. Final implementation, integration, testing, and project decisions were completed manually.

The extent of AI/LLM usage is around 15–20%, mainly for code suggestions and improving code readability, specially for Bootstrap implementation.

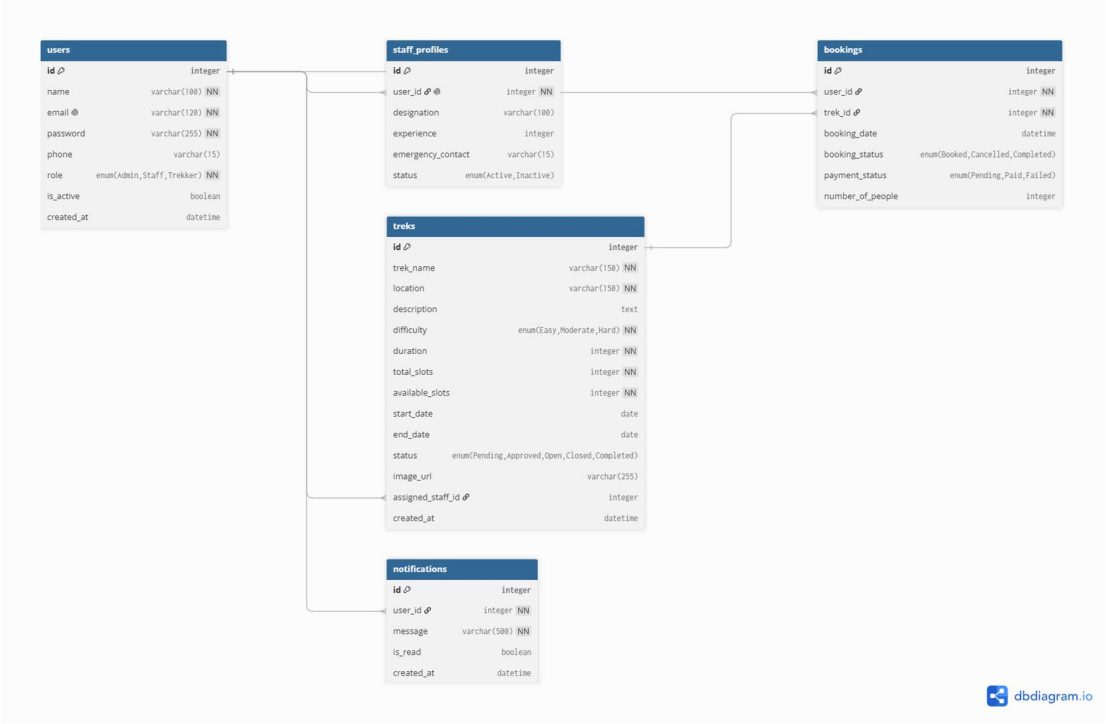
4. Technologies and Frameworks Used

Flask, SQLAlchemy, SQLite, Vue.js, Bootstrap, JWT Authentication, Celery, Redis, HTML, CSS, JavaScript, and REST APIs.

5. Database Schema

Main Entities:
User, Trek, Booking, Notification, Reminder, Review, ExportJob, TrekStaff Assignment.

Relationships:
• Trek → Booking (One-to-Many)
• User → Booking (One-to-Many)
• Trek Staff → Trek (One-to-Many)
• User → Notification (One-to-Many)



6. API Resource Endpoints

/api/auth/login
/api/auth/register
/api/admin/dashboard
/api/admin/treks
/api/staff/dashboard
/api/staff/trek-status
/api/user/dashboard
/api/user/bookings
/api/notifications
/api/reminders
/api/exports

7. Architecture and Features

Backend: Flask + SQLAlchemy
Frontend: Vue.js SPA

Features Implemented:
✔ Authentication and Role-Based Access
✔ Admin Dashboard and Trek Management
✔ Trek Staff Dashboard and Trek Operations
✔ User Dashboard and Trek Booking System
✔ Booking History and Tracking
✔ Notifications and Reminders
✔ CSV/HTML Export Functionality
✔ Scheduled Background Jobs using Celery

Video Presentation Drive Link:

https://drive.google.com/file/d/1AOXZdtWJJU4lYYsbumtlvrsncA423GBl/view?usp=sharing